

# Deepak Mallampati

## APPLIED AI / ML ENGINEER - LLM DEPLOYMENT

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/Deepak0070 |  
Kent, OH

### PROFESSIONAL SUMMARY

Applied AI / ML engineer with 4+ years deploying production LLM systems into customer-facing settings across regulated and enterprise domains. Hands-on across the deployment lifecycle: advanced RAG over large document corpora (cross-encoder re-ranking, ~35% retrieval precision gains), agentic workflows on a conductor-subagent framework (42% less token use), and LLM fine-tuning (Qwen3-27B via 4-bit QLoRA on H100; LoRA/QLoRA/PEFT). Strong on the concepts and algorithms underlying ML and LLMs, with a PyTorch / Hugging Face / LangChain stack and an evaluation discipline (RAGAS, BERTScore, custom metrics) that gates releases. Ships open source (MangaLens, live on the Chrome Web Store). Comfortable explaining complex systems to technical and non-technical stakeholders. US-based, F-1 OPT (open to relocation / sponsorship).

### SKILLS

Python + LLM Apps • Fine-Tuning (QLoRA/PEFT) • Advanced RAG • Agentic Workflows • LangChain + Vector DBs •  
PyTorch / Deep Learning • LLM Evaluation • Customer Deployment • Open Source • FastAPI Services

**Languages:** Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++

**AI / ML:** PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8

**LLM / GenAI:** RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

**Backend:** FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright

**Frontend:** React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

**Data:** PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake

**DevOps:** Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)

**Domains:** Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal

### PROFESSIONAL EXPERIENCE

#### Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability ~**25%**, hallucinations cut >**30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall +**15-20%** on high-risk categories, precision held >**90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream instability -**40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

## Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP (contracts, nominations, allocations, invoicing); query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported RBAC and audit logging for financial modules in a regulated environment.

## Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask API** with a lightweight web interface for non-technical staff.

## PROJECTS

---

### Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies.

Audit-ready explainable risk scoring.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

### Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers: Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for worst-case failure isolation. 29 sites supported; 7-day cache; privacy policy and narrowed permissions.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Ollama

### Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt-transformation layer, grounded image generation. **~30%** contextual alignment gain, **~25-30%** first-pass image success.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

## RESEARCH

---

### Kent State University

Oct 2023 - Jan 2024

ML & Generative AI Research Assistant

Kent, OH - Privacy-Preserving ML, Healthcare Data

- Built a **privacy-preserving ML pipeline** on clinical hospital-readmission data (k-anonymity  $k=3$ , l-diversity  $l=2$ , Laplace differential privacy); reduced re-identification risk **3.38% to 0.32%** while retaining **99%** of baseline predictive performance (RF AUC 0.689 vs 0.696).
- Designed a composite **privacy-utility scoring framework** benchmarking six anonymization configurations to identify the optimal operating point.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper.

EDUCATION

|                                                                                                                                                          |                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Master of Science, Computer Science - Kent State University</b><br>Focus: Applied Machine Learning, NLP, Database Systems.                            | <i>Aug 2023 - May 2025</i> |
| <b>Bachelor of Technology, Computer Science - KL University</b><br>Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives). | <i>Jun 2019 - May 2023</i> |

CERTIFICATIONS

|                                                                  |             |
|------------------------------------------------------------------|-------------|
| Deep Learning Specialization - <i>DeepLearning.AI (Coursera)</i> | <i>2024</i> |
|------------------------------------------------------------------|-------------|